

# Physics-Informed Joint-Embedding Predictive Architecture (JEPA) Triad

Representation-space latent predictive dynamics constrained by fundamental physical conservation invariants

## (A) Observation Drowning vs. JEPA

### Observation-Space Baselines

TimesNet / TranAD / AnomalyTransf.

$$\min_{\theta} \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \text{ in raw sensor domain}$$

- Forces networks to fit high-frequency stochastic sensor noise & benign jitter.
- High reconstruction residual  $\neq$  true anomaly.
- Generates  $> 1,000,000$  false alarms in telemetry.

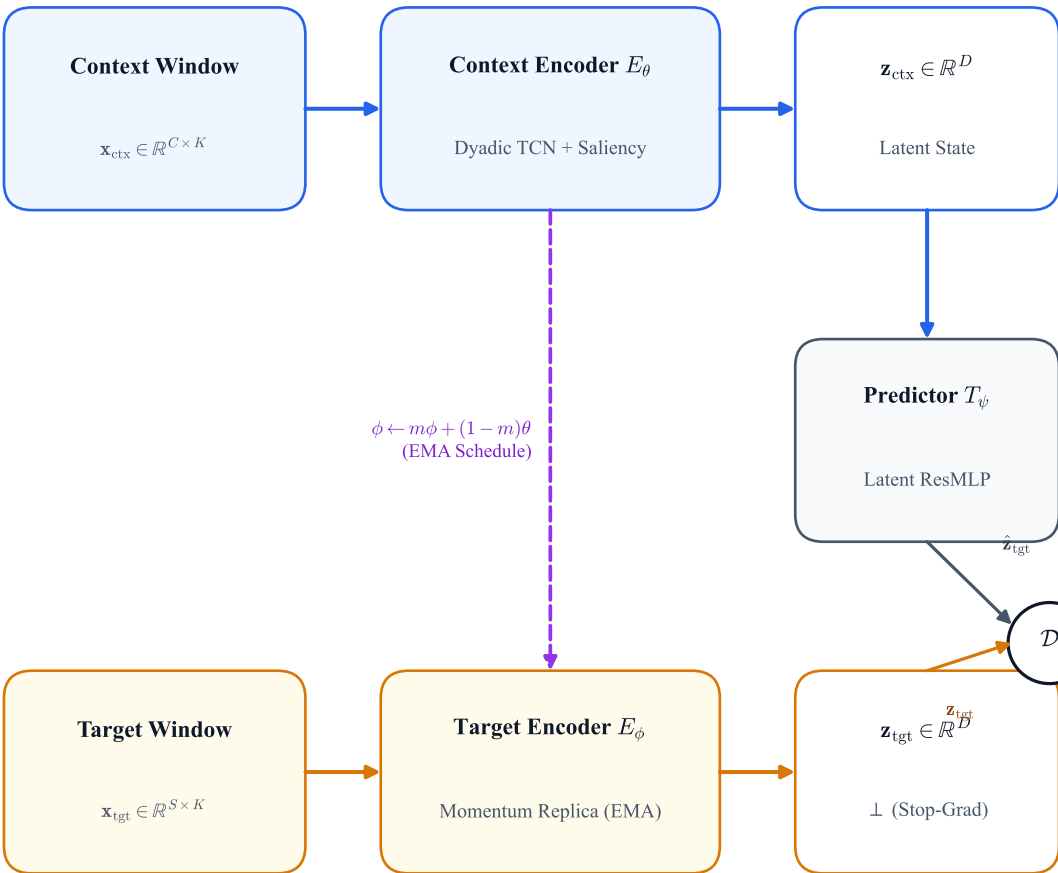
### Proposed Latent JEPA Triad

Operator Entropy / Reynolds / Potential Flow

$$\min_{\theta, \psi} \mathcal{D}(T_{\psi}(\mathbf{z}_{\text{ctx}}), \mathbf{z}_{\text{tgt}}) \text{ in } \mathbb{R}^D$$

- Predicts abstract invariant dynamical state.
- Discards pixel/point stochastic noise.
- Immune to representation collapse via non-contrastive physical conservation heads.

## (B) Unified Predictive Latent Architecture



## (C) Physics Heads & Tail Calibration

### 1. Operator-Entropy JEPA

Koopman Dynamic Spectral Entropy (Unitary Spectrum)

$$\mathcal{K}_{\mathbf{z}} \mathbf{z}_t \approx \mathbf{z}_{t+1}, \quad \mathcal{H}_K = -\sum_i \lambda_i \ln \lambda_i$$

### 2. Reynolds-Stress JEPA

Turbulent Momentum Transfer & Shear Divergence

$$\mathbf{u} = \bar{\mathbf{u}} + \mathbf{u}', \quad \mathbf{R} = \overline{\mathbf{u}' \otimes \mathbf{u}'} \quad (\nabla \cdot \mathbf{R} \neq 0)$$

### 3. Potential-Flow JEPA

Conservative Irrotational Potential Field

$$\mathbf{v} = \nabla \Phi, \quad \nabla \times \mathbf{v} = 0 \quad (\text{Vorticity-Free Conservation})$$

### Pure Inductive EVT & Streaming SPOT

Kurtosis-adaptive Generalized Pareto distribution tail fitting

$$\tau(t) = \text{Med}(t) + (\tau_0 - \text{Med}_0) \cdot \frac{\text{IQR}(t)}{\text{IQR}_0}$$